

39.1 Building a model

A generalized linear model has three parts that can be chosen separately: a response family, which fixes the variance function $V(\mu)$; a link g ; and a linear predictor $\eta = \mathbf{x}^\top \boldsymbol{\beta}$. Chapter 34 fits whichever combination is handed to it; this section is about how the combination is chosen, and how honest one can be afterwards.

39.1.1 The family follows the variance

Two things are known about a response before any model is fitted: what values it can take, and how its spread changes with its level. The first rules out most families at once — a count is not negative, a proportion is bounded, a duration is positive. The second separates those that remain, because within the exponential dispersion family the variance function determines the family (Exercise (B2)). For the families of Definition 34.1.2 the variance is a power of the mean,

$$\text{Var}(Y_i) = \phi \mu_i^\zeta / w_i, \quad (39.1.1)$$

with $\zeta = 0$ for the normal, 1 for the Poisson, 2 for the gamma and 3 for the inverse Gaussian; in between are the Tweedie families of Exercise (C2). So “which family?” is largely the question “which ζ ?”, and with replicates it can be answered before any model is fitted. A second choice arises at once: the scale on which the systematic part is additive. Transforming the response and transforming the mean are different operations (Exercise (B1)), and models on different response scales are not comparable by their likelihoods unless the change of variable is accounted for. The proposition collects the three facts that make these choices more than a matter of taste.

Proposition 39.1.1 — Three facts about building a generalized linear model

- (a) (*Reading the variance function off replicates.*) Suppose the observations fall into $G \geq 2$ groups, group g holding $m_g \geq 2$ independent replicates with mean $\mu_g > 0$, variance $\phi \mu_g^\zeta$ and finite fourth moment. Let $\bar{y}_g$ be the mean of the replicates in group g , let s_g^2 be their variance with divisor $m_g - 1$, and let $\hat{\zeta}$ be the slope of the weighted least squares regression of $\log s_g^2$ on $\log \bar{y}_g$ with any fixed positive weights c_g . If the $\log \mu_g$ are not all equal, then $\hat{\zeta} \rightarrow \zeta$ in probability as $\min_g m_g \rightarrow \infty$ with G , the μ_g and ϕ fixed.
- (b) (*Additivity is a property of the link, not of the data.*) Let $g(\mu(x_1, x_2)) = a(x_1) + b(x_2)$ on $I_1 \times I_2$, with a and b continuous and nonconstant, and let J be the interval of values taken by $a + b$. Let $\tilde{g}$ be a second link and $s = \tilde{g} \circ g^{-1}$, twice differentiable on J . Then $\tilde{g}(\mu(x_1, x_2))$ is again a sum of a function of x_1 and a function of x_2 if and only if s is affine on J .
- (c) (*Comparing models across response scales.*) Let t be strictly increasing and continuously differentiable with $t' > 0$, and let a model for the transformed responses $\tilde{y}_i = t(y_i)$ have maximized log-likelihood $\tilde{\ell}$ and d estimated parameters. As a model for $\mathbf{y}$ it has maximized log-likelihood $\tilde{\ell} + \sum_i \log t'(y_i)$, so its AIC on the scale of $\mathbf{y}$ is

$$\text{AIC} = -2\tilde{\ell} + 2d - 2 \sum_{i=1}^n \log t'(y_i). \quad (39.1.2)$$

Proof.

- (a) With m_g replicates of finite fourth moment, $\bar{y}_g \rightarrow \mu_g$ and $s_g^2 \rightarrow \phi \mu_g^\zeta$ in probability as $m_g \rightarrow \infty$, the second because $\text{Var}(s_g^2) \rightarrow 0$ when the fourth moment is finite. Both limits are positive, and $\log$ is continuous there, so $L_g = \log \bar{y}_g \rightarrow \log \mu_g$ and $V_g = \log s_g^2 \rightarrow \log \phi + \zeta \log \mu_g$ in probability. The weighted slope

$$\hat{\zeta} = \frac{\sum_g c_g (L_g - \bar{L}_c)(V_g - \bar{V}_c)}{\sum_g c_g (L_g - \bar{L}_c)^2}, \quad \bar{L}_c = \frac{\sum_g c_g L_g}{\sum_g c_g},$$

is a continuous function of $(L_1, V_1, \dots, L_G, V_G)$ wherever the denominator is nonzero. At the limit point the denominator is $\sum_g c_g (\log \mu_g - \log \mu_c)^2 > 0$, because the $\log \mu_g$ are not all equal, and the numerator is ζ times the same quantity. The continuous mapping theorem gives $\hat{\zeta} \rightarrow \zeta$.

- (b) If $s(u) = \alpha + \beta u$ then $\tilde{g}(\mu) = \alpha + \beta a(x_1) + \beta b(x_2)$, a sum. Conversely suppose $\tilde{g}(\mu(x_1, x_2)) = A(x_1) + B(x_2)$. If $a(x_1) = a(x'_1)$ then $\mu(x_1, x_2) = \mu(x'_1, x_2)$ for every x_2 , so $A(x_1) = A(x'_1)$; hence $A^*(u) = A(x_1)$ is well defined on the range U of a , and likewise B^* on the range V of b . Both are nondegenerate intervals, being continuous images of intervals under nonconstant maps, and

$$s(u + v) = A^*(u) + B^*(v), \quad u \in U, v \in V.$$

Fix u_0, v_0 interior to U, V . For any other such u, v the four terms cancel:

$$s(u + v) - s(u_0 + v) - s(u + v_0) + s(u_0 + v_0) = 0.$$

Differentiating in u and then in v gives $s''(u + v) = 0$. As $u + v$ ranges over an open subinterval of J whose closure is J , and s'' is continuous, $s'' \equiv 0$: s is affine.

- (c) If $\tilde{Y} = t(Y)$ has density $f_{\tilde{Y}}$, then Y has density $f_Y(y) = f_{\tilde{Y}}(t(y)) t'(y)$, because t is a strictly increasing differentiable bijection onto its image. Summing logs over independent observations, $\ell(\cdot; \mathbf{y}) = \tilde{\ell}(\cdot; \tilde{\mathbf{y}}) + \sum_i \log t'(y_i)$ for every parameter value, so the maximizers coincide and the maximized values differ by the stated constant. Substituting into $\text{AIC} = -2\ell + 2d$ gives [Eq. 39.1.2](#). ■

Part (c) is worth stating because the omitted term is enormous. The AIC that software prints for a normal linear model fitted to $\log \mathbf{y}$ compares it with other models *for* $\log \mathbf{y}$; against a gamma model for $\mathbf{y}$ the comparison is meaningless until $2 \sum_i \log y_i$, a sum of order n , is added back.

Warning. The same applies to cross-validation, to the deviance and to any other criterion built on a likelihood. A predictive score is a score *for a particular quantity on a particular scale*; predicting $\log y$ well and predicting y well are different achievements, and the second is usually what was wanted.

39.1.2 A worked choice

Example. (How long does a strike last)

The durations of 62 contract strikes in United States manufacturing, from one day to 216, are recorded with a measure of unanticipated industrial production at the time. Example (Strike durations) read the same durations as counts and asked only for a variance function; here they are read as what they are, positive and continuous, and the question is which family to use. The response is strongly right-skewed, so a normal linear model is out; which of the positive families fits the mean–variance relationship?

The covariate takes only 9 distinct values, so the data arrive already in replicate groups, of sizes 2 to 18. The weighted regression of Proposition 39.1.1(a) gives $\hat{\zeta} = 1.606$ with a standard error of 0.434, computed from the exact normal-theory variance $\text{Var}(\log s_g^2) = \psi'\{(m_g - 1)/2\}$ — the familiar $2/(m_g - 1)$ is only its large-group limit, and understates it badly for the two groups of size two. Panel (a) of Figure 39.1.1 shows the nine points with the fitted line and the Poisson and gamma references: $\zeta = 2$ is 0.9 standard errors away and $\zeta = 1$ is 1.4, which is weak evidence for the gamma and against anything Poisson-like.

Fit three models with a log link and the same linear predictor, maximizing each over its dispersion parameter as well as over β , and count $d = 3$ for all three. AIC on the scale of the durations is 585.4 for the gamma, 590.9 for the log-normal and 603.3 for the inverse Gaussian. The log-normal figure is the AIC of the least squares fit to $\log \mathbf{y}$, 206.8, plus the Jacobian term $2 \sum_i \log y_i = 384.1$ of Eq. 39.1.2. Without that term the log-normal appears to beat the gamma by 378.7: not a close call, and in the wrong direction, because the uncorrected number answers a different question. Leave-one-out cross-validation with the log score puts the three in the same order, at -4.7192 , -4.7713 and -4.8664 per observation.

The gamma slope is to the last decimal the estimate that Example (Strike durations) obtained from $V(\mu) = \mu^2$ alone: the variance function fixes the estimating equations, and choosing a family on top of it buys a likelihood, not a different fit. The maximum likelihood shape of the gamma fit is $\hat{\nu} = 1.003$, so the durations are close to exponential given the covariate, which is what $\zeta = 2$ with a coefficient of variation near one means. (Software reports instead the moment estimate from the Pearson dispersion $\hat{\phi} = 0.941$, namely 1.063; the two agree here, and for the inverse Gaussian they do not.) The fits also disagree about what they estimate: at the mean covariate value the gamma model puts the *mean* duration at 39.4 days, while the log-normal model puts the *median* at 22.2 and, after the retransformation correction of Proposition 22.5.1, the mean at 46.7.

```
import numpy as np
import statsmodels.api as sm
strikes = sm.datasets.strikes.load_pandas().data      # 62 contract strikes, public
    domain
y = strikes["duration"].to_numpy(float)              # days
x = strikes["iproduct"].to_numpy(float)              # unanticipated industrial
    production

groups = strikes.groupby("iproduct")["duration"].agg(["count", "mean", "var"])
groups = groups[groups["count"] >= 2]                # nine groups, sizes 2 to 18
w = groups["count"].to_numpy() - 1.0                 # weights: m_g - 1 degrees of
    freedom
```

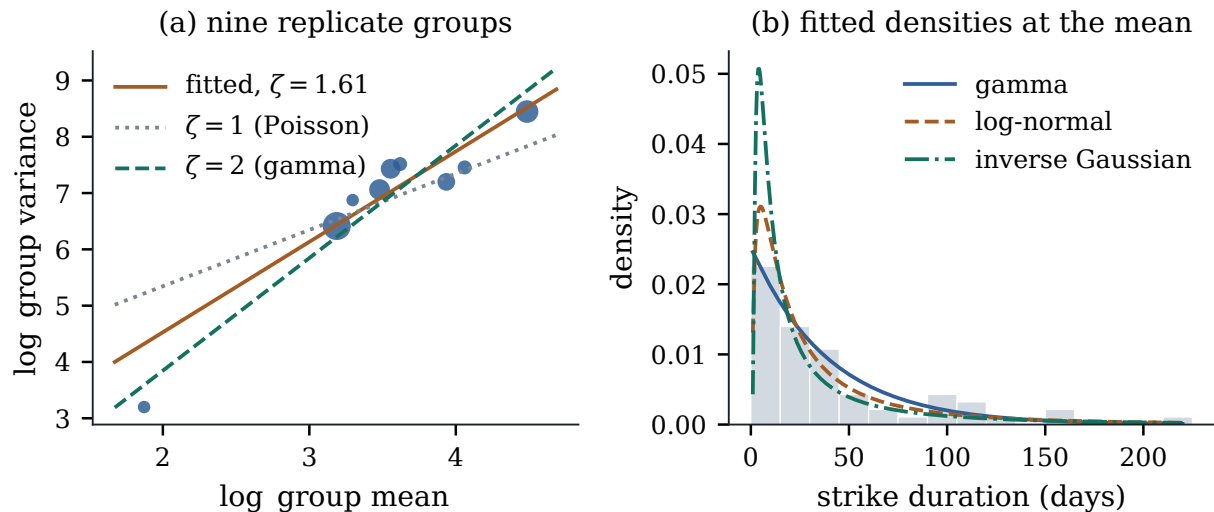


Figure 39.1.1: Choosing a family for the strike durations. (a) Log group variance against log group mean, point size proportional to group size; the fitted slope estimates ζ in Eq. 39.1.1, and the reference lines have the Poisson and gamma slopes. (b) The three fitted densities at the mean covariate value, over a histogram of the durations.

```
L = np.column_stack([np.ones(len(groups)), np.log(groups["mean"])])
zeta_fit = np.linalg.solve(L.T @ (w[:, None] * L), L.T @ (w * np.log(groups["var"])))
)

print(groups.round(2))
print(f"variance-function power  zeta = {zeta_fit[1]:.3f}")
```

[Run this code online](#)

```
from scipy import optimize, special, stats
X = np.column_stack([np.ones(len(y)), x])
n, p = X.shape
d = p + 1                                     # slopes plus one dispersion
parameter

gam = sm.GLM(y, X, family=sm.families.Gamma(sm.families.links.Log())).fit()
igauss = sm.GLM(y, X, family=sm.families.InverseGaussian(sm.families.links.Log())).fit()
lognorm = sm.OLS(np.log(y), X).fit()          # a normal linear model for
log y

def gamma_loglik(yy, mu, nu):
    return np.sum(stats.gamma.logpdf(yy, nu, scale=mu / nu))

def gamma_shape(yy, mu):
    """Maximum likelihood shape: the root of log nu - digamma(nu) + mean log(y/mu)."""
    c = np.mean(np.log(yy / mu))
    return optimize.brentq(lambda v: np.log(v) - special.digamma(v) + c, 1e-4, 1e4)

def igauss_lambda(yy, mu):
    """Maximum likelihood precision of the inverse Gaussian, in closed form."""
```

```

    return 1.0 / np.mean((yy - mu) ** 2 / (mu**2 * yy))

nu_hat = gamma_shape(y, gam.fittedvalues)
lam_hat = igauss_lambda(y, igauss.fittedvalues)
ll_gam = gamma_loglik(y, gam.fittedvalues, nu_hat)
ll_ig = np.sum(stats.invgauss.logpdf(y, igauss.fittedvalues / lam_hat, scale=lam_hat
    ))
sigma_ml = np.sqrt(lognorm.ssr / n)                                # the maximizing sigma,
    divisor n
ll_ln = np.sum(stats.norm.logpdf(np.log(y), lognorm.fittedvalues, sigma_ml) - np.log
    (y))

for name, ll in [("gamma, log link", ll_gam), ("inverse Gaussian", ll_ig),
    ("log-normal (on the y scale)", ll_ln)]:
    print(f"{name:28s} loglik {ll:9.2f}    AIC {-2 * ll + 2 * d:9.2f}")
print(f"gamma:  slope {gam.params[1]:8.3f},  maximum likelihood shape {nu_hat:.3f}")

```

[Run this code online](#)

39.1.3 Links, scales and interactions

With the family fixed, the link decides what “additive” means. The canonical link (Definition 34.2.2) keeps the log-likelihood concave (Theorem 34.3.3) and makes observed and expected information agree (Theorem 34.3.1(d)); neither is a reason to prefer it when its range constrains β . The gamma’s canonical link is the reciprocal, which forces $\mathbf{x}^\top \beta > 0$ and makes every coefficient a statement about $1/\mu$; the log link has no constraint, keeps the concavity (Exercise (C1)), and makes each coefficient a multiplicative effect on the mean. Almost all gamma regressions use the log link, for interpretation rather than fit.

For the covariates, Theorem 22.3.2 transfers verbatim with the linear predictor in place of the mean: to ask whether x_j should enter as x_j^λ , add $x_j \log x_j$ and test its coefficient. Centring matters for a different reason. Under a nonlinear link the intercept is the linear predictor at $\mathbf{x} = \mathbf{0}$, a point no observation resembles when the covariates are uncentred; centring each at a representative value makes $e^{\hat{\beta}_0}$ the fitted mean of a typical case.

Interactions are where the link stops being cosmetic. A two-factor Poisson model with a log link and no interaction says the factors multiply; with an identity link it says they add; by Proposition 39.1.1(b) at most one of these is the additive model of Definition 16.2.1. “No interaction” is therefore never a property of the response alone, and removing an interaction by changing the link is a legitimate simplification, provided the link is reported as part of the finding and was not chosen by looking at the interaction test (Proposition 29.5.1). On the log scale an interaction coefficient is a ratio of ratios: with binary x_1, x_2 and $\log \mu = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_{12} x_1 x_2$, $e^{\beta_{12}}$ is the factor by which the effect of x_1 changes when x_2 switches on; on the logit scale it is a ratio of odds ratios, harder to think about because odds ratios do not collapse over omitted covariates (Example (The odds ratio does not collapse)). Offsets need no new theory (Proposition 37.2.1): a known exposure enters with coefficient fixed at one, and the model describes a rate.

39.1.4 Comparing and choosing

Because a generalized linear model is fitted by maximum likelihood, everything in [Chapter 29](#) applies: AIC is $-2\ell(\hat{\beta}) + 2d$ with $d = p$ when ϕ is known and $p + 1$ when it is estimated, BIC replaces 2 by $\log n$ (Definition 29.2.2), and their different targets (Proposition 29.2.4) are unchanged. Two cautions are specific here. First, by [Proposition 39.1.1\(c\)](#) the likelihoods compared must be for the same random variable, and each must be maximized over the dispersion as well as over β . Second, a quasi-likelihood model has no likelihood, so no AIC; the substitute is the quasi-likelihood information criterion built from the independence-model quasi-likelihood and a sandwich penalty, which [Chapter 40](#) defines. The dispersion estimate is not a likelihood quantity either (Proposition 38.3.2), which is why an overdispersed fit is compared by quasi- F statistics.

Cross-validation needs a loss, and the natural one is the log score: hold out an observation, fit to the rest, record the predictive log density of what was held out. That is Definition 29.3.1 with the deviance as loss, so Proposition 29.3.1 applies. For a $\phi = 1$ family the log score and the deviance rank models identically; with a dispersion parameter they need not, because the deviance ignores the term in ϕ .

Idea. Three questions, three tools. *Which family?* The support and the mean–variance relationship. *Which link?* What should be additive, checked afterwards ([Section 39.2](#)). *Which terms?* A criterion, plus honesty about what the search cost.

39.1.5 What the search costs

Every warning of [Section 29.5](#) carries over unchanged, because none used normality: after a search the standard errors are too small, the p -values too small and the intervals too short (Proposition 29.5.1). Two remarks are specific here. A drop in deviance is approximately $\chi^2(q)$ under the smaller model (Theorem 34.5.5), so a maximum over many candidate terms is not, and the largest always looks significant; and with small n or sparse cells that approximation is itself poor (Example (The Hauck–Donner effect), Proposition 35.4.1), so the nominal levels being abused were never right to begin with. Sample splitting (Proposition 29.5.2) and simultaneous inference over the whole search (Proposition 29.5.3) are the honest ways out.

39.1.6 Exercises

39.1.6.1 A. Check your understanding

Exercise A1.

A colleague fits a normal linear model to $\sqrt{y}$ and a Poisson log-linear model to y , and compares the two AIC values printed by the software. What is the correction term of [Proposition 39.1.1\(c\)](#) here, and in which direction does omitting it push the comparison?

Solution. With $t(y) = \sqrt{y}$, $t'(y) = 1/(2\sqrt{y})$, so the correction term is

$$-2 \sum_i \log t'(y_i) = 2 \sum_i \log(2\sqrt{y_i}) = 2n \log 2 + \sum_i \log y_i,$$

and the AIC of the square-root model on the scale of $\mathbf{y}$ is its printed value plus that amount. It is positive whenever $\prod_i 4y_i > 1$, as for counts of any reasonable size it is, so the printed comparison flatters the square-root model. The Poisson model needs no correction, being already a model for $\mathbf{y}$ — though it is a mass function rather than a density, which makes the comparison dubious anyway.

Exercise A2.

A two-by-two table of counts has fitted means $\mu_{11} = 10$, $\mu_{12} = 20$, $\mu_{21} = 15$ and $\mu_{22} = 30$. Is there an interaction on the log scale? On the identity scale? Reconcile the two answers with [Proposition 39.1.1\(b\)](#).

Solution. On the log scale, $\log \mu_{11} + \log \mu_{22} - \log \mu_{12} - \log \mu_{21} = \log(10 \cdot 30 / (20 \cdot 15)) = \log 1 = 0$: no interaction. On the identity scale, $10 + 30 - 20 - 15 = 5 \neq 0$: there is one. The two links differ by the map $s = \exp$, which is not affine, so [Proposition 39.1.1\(b\)](#) says there is no reason for both to be additive, and here only one is.

39.1.6.2 B. Practice

Exercise B1.

The canonical link of the gamma family is $g(\mu) = -1/\mu$ (Example (The gamma and its constant coefficient of variation)). Show that with this link and a model matrix containing an intercept, the parameter space $\{\boldsymbol{\beta} : \mathbf{x}_{(i)}^\top \boldsymbol{\beta} < 0 \text{ for all } i\}$ is an open convex cone, and that the maximum likelihood estimate may fail to exist by running to its boundary. Contrast with the log link.

Solution. The set is an intersection of n open half-spaces, hence open and convex, and a cone because each constraint is homogeneous. The log-likelihood tends to $-\infty$ as any $\mathbf{x}_{(i)}^\top \boldsymbol{\beta} \rightarrow 0^-$, since $\mu_i \rightarrow \infty$; but the supremum may be approached only as $\|\boldsymbol{\beta}\| \rightarrow \infty$ inside the cone, in which case no maximizer exists, exactly as in Theorem 35.3.1. With the log link the parameter space is all of $\mathbb{R}^p$ and the log-likelihood is concave (Exercise (C1)), so the only obstacle is the analogue of separation.

Exercise B2.

Show that for a family with $\phi = 1$ the leave-one-out log score and the leave-one-out deviance differ by a constant that does not depend on the model, so the two rank models identically. Show that this fails when ϕ is estimated.

39.1.6.3 C. Going deeper

Exercise C1.

Extend [Proposition 39.1.1\(b\)](#) to three covariates: if $g(\mu) = a_1(x_1) + a_2(x_2) + a_3(x_3)$ with all three nonconstant and continuous, and $\tilde{g}(\mu)$ is also a sum of three functions of the separate covariates, must $s = \tilde{g} \circ g^{-1}$ be affine? Does the conclusion change if only two of the three are required to separate?

Exercise C2.

Let the true link belong to the power family $g_\lambda(\mu) = (\mu^\lambda - 1)/\lambda$, which has $g_0 = \log$, and let the model actually fitted use the log link. Following the constructed-variable argument of Proposition 22.2.4, expand g_λ about $\lambda = 0$ to show that a one degree of freedom test of $\lambda = 0$ is obtained by adding a multiple of $\hat{\eta}^2$ to the linear predictor, and identify the constructed variable and the multiple exactly.